Prosthetic Training: Upper Limb

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KEYWORDS

• Occupational therapy • Prosthetic training • Rehabilitation • Upper limb
• Prosthetics

KEY POINTS

- A collaborative approach is essential to successful rehabilitation.
- The choice of componentry determines the training needs.
- Prosthetic component manufacturers are a useful resource.
- Managing expectations during therapy is crucial in helping minimize client frustration.
- Mastering skills before proceeding to the next set of skills during training is vital for improved client outcomes.

INTRODUCTION

For the individual who receives an arm prosthesis, occupational therapy is a critical piece of the overall rehabilitation puzzle. An occupational therapist (OT) will help the individual learn to use their prosthesis; use it for activities of daily living (ADL); and, it is hoped, incorporate it into their everyday life. But an OT can only do so much.

If the prosthesis does not fit, is poorly designed, has components that are inappropriate, has components that are assembled incorrectly, and/or is improperly programed, then the OT will struggle greatly or will be unable to train the individual how to use their prosthesis properly. This scenario could lead to frustration for the OT and especially the individual with limb loss. Without proper knowledge of what components are appropriate or how a prosthesis is supposed to fit, individuals might think it is their fault and that they need to try harder and practice more, when ultimately this will lead nowhere. The OT, in turn, may wonder what they are doing wrong. It is imperative that there is constant collaboration and frequent communication between the certified prosthetist (CP) and the OT for the best overall outcome for individuals with an arm prosthesis.

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For the purposes of this text, this article solely focuses on adult upper limb prosthetic training. Pediatric prosthetic training will vary significantly secondary to the nature of the limb loss/limb difference. Parental involvement is a crucial factor, and training typically happens in phases according to the various developmental milestones and stages of life. Many resources are available for pediatric prosthetic training.[1–4]

TYPES OF PROSTHESES: AN OT'S PERSPECTIVE

The OT essentially needs to know that there are 5 types of prostheses and the various terms that might be used to describe the same type of prosthesis:

1. Cosmetic or passive functional
2. Body powered or cable driven
3. Electrically powered (myoelectric or switch controlled)
4. Hybrid (combination of body powered and electric)
5. Activity specific (designed for a specific task, such as swimming or showering)

But an OT should be able to rely on the CP for more detailed information regarding the various types of prostheses available for individuals with upper limb loss. Please see the article by Kistenberg elsewhere in this issue.

To properly train an individual with an arm prosthesis, the OT must understand the following:

- What type of prosthesis their patient has
- What componentry is on the prosthesis
- If electric, how the components are programed

The OT and CP should communicate on an ongoing and regular basis to discuss any questions regarding the prosthesis as well as any problems, issues, or barriers that might impede training.

WORKING CLOSELY WITH THE CP AND PROSTHETIC COMPONENT MANUFACTURER

The OT should form a close working relationship with the CP and even attend prosthetic appointments with their client if possible and when necessary. The same should go for the CP attending OT appointments because this will only help both professionals understand what it is the other is looking for and more closely appreciate each other's roles and responsibilities because some items may overlap or one professional might have a resource that the other was unaware of.

The CP has typically formed a relationship with the prosthetic component manufacturer, which is crucial with today's evolving and rapidly advancing technology. The prosthetic manufacturers offer instructional courses and training on their specific components and technology and, in some instances, require the CP to bring and fit a client in order to become a certified provider of that particular technology. The prosthetic manufacturers encourage the CPs to bring OTs to courses because this helps foster the relationship and enhances the OT's knowledge base. The OT is not required to understand all of the intricate details of the components, technology, and programing; however, the more knowledge acquired, the better they are able to train individuals how to use their prosthesis and use it in their ADL.

GOALS AND TIME FRAMES

The time from an individual being casted for a prosthesis to incorporating it into their daily life varies greatly. General time frames for prosthetic training rehabilitation are available[5]; but it is important to keep in mind that the rehabilitation time frames are dependent on many factors, which can heavily influence the time spent in therapy. Some of those factors include the following:

- Nature of limb loss/limb difference (acquired vs congenital)
- Concomitant injuries
- Level of limb loss
- Number of limbs affected
- Type of prosthesis
- Componentry used
- Programing
- Previous prosthetic wear/use
- Previous therapy
- Experience level of prosthetist
- Motivation of client
- Client cognition
- Experience level of therapist
- Availability of client
- Availability of prosthetist
- Insurance

In some cases, individuals may receive a preparatory prosthesis[6] within a day or two (typically referred to as an expedited fitting); in other cases, it may be months before they receive a definitive or finalized prosthesis. Whether an individual receives a preparatory or definitive prosthesis is up to the prosthetist. It is recommended that the OT be involved from the initial stages of the prosthetic fabrication process and begin training with a preparatory prosthesis. The OT will be able to put the client in various therapy situations, which will help determine if any modifications need to be made before the CP finalizes the prosthesis. It is easier to modify a preparatory prosthesis than a definitive prosthesis. Once training has begun, it could take anywhere from a few days to several months in order for the client to feel proficient with their prosthesis. Again, the time frame depends on the factors mentioned earlier. Managing expectations is important. The proliferation of feel-good stories in the media and online can cause/create a significant disconnect between a client's expectations and realistic expectations. Early prosthetic training is feasible if there are no outstanding preprosthetic rehabilitation areas to be addressed.

The goals of upper limb prosthetic training should be client centered and meaningful. The OT and family members may also have certain goals in mind for the client that will help them be more independent with basic self-care tasks; but if it is not meaningful to the client, then it may not be appropriate to address it in therapy because it could lead to frustration and a battle of wills. That does not mean the client will not find those tasks meaningful or important in the future, therapy tends to be a process that occurs in phases, and those tasks can always be addressed at a later time. All that being said, it is important to discuss with the client the need for independence with self-care activities in the event someone is not around to help. Some common goals the OT should keep in mind include the following:

- Independence with don/doff of the prosthesis
- Perform basic self-care tasks independently with the prosthesis or adaptive equipment (feeding, bathing, toileting, dressing)

- Perform ADL independently with the prosthesis or adaptive equipment (feeding, meal preparation, grooming, home management)
- Engage in leisure activities/hobbies with the prosthesis or adaptive equipment

OCCUPATIONAL THERAPY INITIAL ASSESSMENT

It would be ideal if the same OT performed both the preprosthetic evaluation and prosthetic training evaluation, but this may not always be the case. The preprosthetic phase of rehabilitation should have prepared the client and his or her body/limb to wear, tolerate, and use a prosthesis, which will ultimately make the prosthetic training phase much smoother and faster.

If an individual has not received any preprosthetic rehabilitation, it may make the prosthetic training phase difficult for the following reasons:

- Range of Motion (ROM) deficits
- Strength deficits
- Scar tissue adhesions
- Edema
- Hypersensitivity
- Pain issues

For example, adherent scar tissue will affect the individual's ability to wear a socket without skin breakdown. Edema or volume fluctuation in the residual limb will affect the fit of the socket and ultimately the client's comfort and/or control ROM or strength deficits will affect the individual's ability to control the various components on their prosthesis. These areas must be addressed before prosthetic training can begin, and changes to the prosthesis may be necessary once these areas have been resolved. This will be a source of frustration for both the OT and especially the client who was expecting to learn how to use their new prosthesis and instead may feel like they took 3 steps back. Please see the article by Klarich and Brueckner elsewhere in this issue.

In any event, the OT initial assessment should paint a very thorough picture of the individual and describe in detail any problem areas in order to justify what needs to be addressed during the prosthetic training phase and identify barriers to training. The areas to be addressed during the assessment should include, but are not limited to the following:

- Cause and onset
- Age, hand dominance
- Other medical issues
- Surgeries
- Medications
- Level of independence with ADL
- ROM
- Strength
- Skin integrity
- Shape of residual limb
- Phantom pain
- General pain
- School and/or work/vocational goals
- Support
- Home environment
- Previous therapies

- Previous prostheses
- Client goals

PROSTHETIC DELIVERY OVERVIEW: AN OT'S PERSPECTIVE

The OT is typically the health care professional that spends the most time with individuals with upper limb loss. Although the CP will give an overview of the componentry, explain the various features, provide education regarding prosthetic wear and care, and demonstrate how to put the prosthesis on and take it off, this might only happen when the client first receives their prosthesis. The OT should be familiar with all of the items discussed during prosthetic delivery so it can be reviewed as necessary during prosthetic training. The OT should review the following areas:

- Prosthesis wear schedule
- Skin checks
- How to take care of and clean their prosthetic socket
- Harness and components
- What hazards to avoid
- What the various components are
- How the components work/operate on their prosthesis
  - Terminal device (TD)
  - Wrist
  - Harness
  - Cables
  - Liner
  - Socket

If the prosthesis is electric, the OT should thoroughly review the various components (TD, wrist, and elbow) and how they are programmed. Once the prosthetic training begins, repetition of this information will help it sink in and make more sense.

The OT should also teach clients how to put their prosthesis, including socks and liners, on and remove it independently. There are various methods for don/doff of arm prostheses. The methods will differ depending on the type of socket, type of prosthesis, personal preference, and the level of limb loss. For someone with a body-powered prosthesis, the methods are similar to putting on or taking off a sweater or a coat. If the individual is wearing a myoelectric prosthesis, they may have a suction socket and require a pull sock to don their arm for an appropriate fit and electrode contact. If an individual is missing more than one limb, adaptive equipment may be necessary. The adaptive equipment can range from a simple work hook mounted to a wall or to a complex dressing tree for individuals with higher-level bilateral limb loss (**Figs. 1** and **2**).

WHAT TO LOOK FOR BEFORE TRAINING BEGINS

If an individual has an electrically powered prosthesis, it is important for the OT to take a few steps back and assess things at a very basic level. This assessment will help the OT determine if there are any problem areas that could impede training and lead to frustration during the actual training. Problem areas can include but are not limited to the following:

- Muscle strength and endurance
- Socket fit
- Electrode placement

Fig. 1. Work Hook with Suction Cup. (*Courtesy of* Shawn Swanson Johnson. Skills for Life 4: Bilateral Upper Limb Loss Workshop.)

- Electrode gain settings
- Incorrect wiring
- Device programing
- Patients' general understanding/comprehension

Because of the advances in prosthetic technology hardware and software over the past 10 to 15 years, it is now possible to virtually assess an individual's ability to operate a prosthesis (Fig. 3).

Fig. 2. Dressing tree. (*Courtesy of* Suzanne Krenek Andrews, TIRR Memorial Hermann.)

Fig. 3. Assessment of prosthesis operation. (*Courtesy of* Ottobock, Minneapolis, MN.)

VIRTUAL EVALUATION AND TRAINING

Before this technology was available, training someone with an electric prosthesis was an inexact science. Clients would put their prosthesis on; try to operate whatever componentry they had (hand, wrist, or elbow); and, if they demonstrated control, then it was time to move onto a few drills and ADL. If they did not demonstrate control, OTs, CPs, and their clients would be left to guess what the possible problems might be. Most of the time, it was assumed that the clients just needed more practice/training. However, if the underlying problem was socket fit, electrode placement, or electrode gain settings, no amount of practice was going to help clients learn how to use their prosthesis. The ability to observe an individual's muscle strength and their ability to control a virtual prosthesis before ever being fit with an actual prosthesis now eliminates most of the guesswork. The OT will need to use a combination of computerized software and hardware that various prosthetic manufacturers offer as well as 2 electrodes with cables. The electrodes should be placed on the bulk of the muscle belly as a starting point and then moved from there to determine the most optimal muscle sites for control. The electrode gain settings should ideally start at a lower level (for example, when using an Otto Bock electrode (Duderstadt, Germany), start at a level between 2–4) because this gives an unadjusted or real view of an individual's muscle strength without any artificial strengthening of the muscles. The optimal muscle sites will depend on the following:

- Strength
- Controllability
- Endurance

In addition, the muscle sites need to fall within the trimlines of the future socket. There is no point evaluating or training muscles that fall outside of the socket area because it will only lead to frustration. Once the electrodes are determined to be properly located, adjustments to electrode gains can be made for easier and optimal control. The electrode gain should not be too high because it will be too sensitive and clients will lack the ability to control slow speeds or could inadvertently operate their wrist. The ability to control slow speeds is necessary to avoid crushing or dropping objects. The electrode gain should not be too low or clients will be unable to generate enough muscle strength to operate any of their prosthetic components. The OT is looking for clients to have natural and easy control. Clients should neither have to struggle or shake to elicit a good muscle contraction nor hold a muscle contraction for a long period of time. Clients with a higher level of limb loss should not have extraneous or exaggerated movements and should be able to isometrically isolate the muscle they want to contract. In order for clients to understand what an easy, natural contraction should feel or look like, the OT can demonstrate on themselves first and then place the electrodes on the clients' sound limb before proceeding to their residual limb. The OT should advise clients to perform the movements and/or training with both arms during the virtual assessment (**Fig. 4**). For example, in order to open the TD, clients with a transradial amputation should extend their wrist and fingers on their sound arm and also imagine doing it with their phantom wrist and fingers at the same time.

Once the muscle sites are selected and the electrodes are adjusted, the OT needs to understand how the clients' prosthetic arm is programmed. There are many different variations of programs to choose from and depending on how the client presents. Factors to consider include the following:

- Muscle strength
- Number of muscles are available for control
- Cognitive ability
- Problem-solving ability
- Level of limb loss

After the CP determines the programing and the OT understands the programing, virtual training can begin. Depending on the virtual software/hardware being used, virtual training can happen in various ways; this type of training will help strengthen the muscles sites, work on endurance, teach clients how to switch between components, and generally build confidence. It typically includes an assessment screen, fun games, and virtual prosthetic component training. On the evaluation screen, the OT can verbally cue for what they want the clients to do (for example, "Try to reach this line quickly with your wrist extensors, which is the red signal, or try to slowly contract your bicep muscle, which is the blue color, and stay under this line."). The games (car maze, manipulating balls) can be a fun way for clients to train and practice without

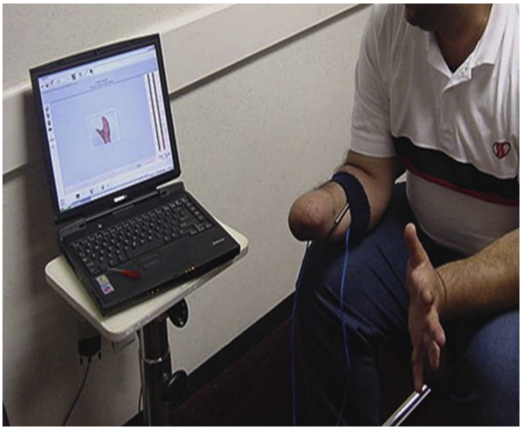

Fig. 4. Virtual training of amputee.

realizing that they are actually working. The virtual prosthetic component screen allows clients to control the various components on a computer screen, which simulates wearing a prosthesis (minus the weight). During this time, clients will learn the various nuances of how their prosthetic arm may be programed or controlled. Clients should also be advised that they do not have to hold a contraction in order to keep the TD open or closed.

Virtual training should be done sitting, standing, and in various planes of the body to demonstrate real-world scenarios; but the electrode cables can make this difficult during the evaluation phase.

If clients are successful with the virtual training, theoretically they should experience success with their actual prosthesis. This virtual training should help reduce the actual prosthetic training treatment time because the clients are already familiar with controlling a virtual prosthesis. The only major differences at this point are the fit and weight of the socket and the ability to do drills and activities while moving about the room. If clients experience problems once they have their prosthesis on, then it is time to reassess and discuss the following with the prosthetist in a collaborative manner:

- Electrode placement
- Socket fit
- Harnessing
- Componentry programing

If the electrode is not in the optimal location, it may cause problems with the controllability of the various electric components. Moving the electrode slightly (either from side to side, down/up, or adjusting the angle) may help clients elicit a stronger or more consistent signal. If the socket is not fitting comfortably and intimately, then the clients' skin may not be making appropriate contact with the electrodes in the socket. Clients will not be able to control their prosthesis consistently or accurately if the socket does not fit. The various devices may move, but it may be because the clients' skin is rubbing along the electrode causing a stray or inadvertent signal not an intentional signal meant to control a device. If clients have to sit or hold their prosthesis on with their other arm, that is a sign of an improper fit. Clear test sockets are extremely helpful during early training to problem solve.

TRAINING DEPENDS ON TYPE OF PROSTHESIS

The stages of prosthetic training are typically the same regardless of the type of prosthesis, but the nature and length of the training will differ depending on the following factors:

- Level of limb loss
- Number of limbs affected
- Type of prosthesis
- Type of components
- If electric, componentry programing

That being said, all types of prosthesis require training and, as mentioned previously, will vary based on the type of prosthesis that clients have been prescribed. For clients with a cosmetic/passive functional prosthesis, training will be minimal and include ways for individuals to stabilize or support objects. The OT will typically have clients stabilize their cosmetic arm on top of a piece of paper for writing or have clients use the cosmetic arm to support a tray of food at a restaurant. For the activity-specific prosthesis or activity-specific TD, training will also be minimal because it is particular to the activity clients want to perform, such as showering, swimming, fishing, weight training, field sports, rock climbing, and so forth. Keep in mind that activity-specific training will most likely be happening at the conclusion of their primary prosthesis training whether it is body powered, electric, or hybrid.

THREE STAGES OF UPPER LIMB PROSTHETIC TRAINING

For the body-powered, electrically powered, or the hybrid types of prosthesis, there are 3 stages of prosthetic training:

- Controls training
- Repetitive drills
- Bimanual functional skill training

These 3 stages are important building blocks for successful prosthetic training. It is not advised to have clients put on a prosthesis and have them attempt to feed or dress themselves on the first visit. This practice will lead to frustration and may cause clients to abandon their prosthesis before proper training has even begun. Clients should master each stage (95%–100% accuracy and consistency) before moving on to the next stage. This progression of training allows the therapist the ability to tease out any problem areas whether it is with the prosthesis and its components, socket or harness fit, or with the patients' physical and/or cognitive ability. If clients are having difficulty with basic operation of their components, then they are certainly going to have difficulty with grasping and releasing objects and performing 2-handed activities. The components should not be moving on their own or operating randomly and intermittently. If there is a problem in any one of the 3 stages, then the therapist can report back to the prosthetist to determine if any modifications or adjustments need to be made before moving on to the next stage. There is a method to the madness, and explaining each of the stages to clients and what to expect during each stage and why they are being done facilitates buy-in and an overall understanding of the process (**Table 1**).

Keep in mind that people complete tasks differently. How an activity is completed will depend on the patients' level of amputation and prosthetic device. Therapists typically learn more from patients than the patients learn from the therapists. The more exposure and experience a therapist has with individuals with limb loss, the more knowledge they will have to pass on to the next person. In addition, the Internet has an abundance of videos with instructional tutorials on how to complete certain tasks. Lastly, it is advisable to introduce current patients to former patients, especially if there is someone who has a similar level of limb loss and prosthetic componentry. It is invaluable for them to be able to talk to someone else who has gone through similar experiences. This communication allows the current patient to see what challenges may lie ahead, how others cope, how others complete certain activities, and so forth.

ELECTRICALLY POWERED COMPONENT SELECTION AND PROGRAMING: HOW IT AFFECTS TRAINING

Electric prosthesis training will differ slightly depending on what type of components are being used and how they are programmed. Many of the prosthetic manufacturers provide training guidelines or informational brochures with their devices or components. These guidelines are extremely helpful in better understanding the technology so the OT can explain to the clients exactly how their componentry works (see **Table 2** for manufacturer information). Also, if possible, the OT should connect themselves to

Table 1
Three stages of prosthetic training

Stage	What Does it Entail?	Purpose and How to Perform	When is it Appropriate and How Long Does it Last?
1. Controls training	• Open and close the terminal device • Rotate and/or flex and extend the wrist • Flex and extend the elbow • Position the elbow • Position the shoulder • Control speeds • Switch between components	Simon says or mirror image of the therapist Teach the clients how to operate all of the various components on their prosthesis before introducing objects to grasp and release Look for clients to achieve consistent and accurate control; prosthesis should do what clients want it to do every time	Initial stage of prosthetic training; can last a few minutes to a few days; should be mastered before moving on to the next stage
2. Repetitive drills	• Foam or wooden therapy peg board • Stacking cones • Clothespin relocation task • Transferring blocks and balls into a bucket • Ring tree • Styrofoam or plastic cup • String beads	Should be done in sitting, standing, and walking; transferring play objects of various sizes and densities to locations of various heights and levels; practice graded grasp and release without dropping or crushing; repeat, repeat, repeat Give clients confidence the prosthesis will do what it is supposed to do in a therapy or practice situation before moving on to a 2-handed task	Initial and middle stages of prosthetic training; can last from a few hours to a few days; should be mastered before moving on to the next stage
3. Bimanual functional skill training	• Twist on/off lids • Pour water into a cup (prosthesis holding cup) • Place a pillow in a pillow case • Pack a suitcase • Wrap a present • Carry a tray • Manage money • Prepare and eat a meal • Many other meaningful bimanual ADL	Engage in tasks clients will encounter on a routine basis; should be meaningful and progress from simple to complex activities Teach clients independence in ADL using their prosthesis; forcing 2-handed situations to minimize stress on unaffected arm	Middle and end stages of prosthetic training; can last a few days to a few weeks; home and work activities should be addressed; may be done in incremental stages as new needs may arise over the course of the clients' prosthetic use

Table 2
Upper limb loss rehabilitation resources

Conferences	Videos	Upper Limb Prosthetic Manufacturers*	Continuing Education Courses	Outcome Measure Information
ISPO World Congress http://www.ispoint.org/events/world-congress	Training the Client with an Electrically Powered Prosthesis Info@UtahArm.com or (888) 696–2767	Fillauer Companies, Inc http://www.fillauercompanies.com/ Hosmer Dorrance Corporation http://hosmer.com/ Motion Control http://www.utaharm.com/	Otto Bock Academy (online tutorials, live classes, interactive webinars) http://academy.ottobockus.com/	Southampton Hand Assessment Procedure
MEC Symposium http://www.unb.ca/conferences/mec/	Skills for Life 2: Bilateral Upper Limb Loss Workshop Proceedings www.resrec.com	Liberating Technologies http://www.liberatingtech.com/	AOTA Approved Provider Program (remote and live courses) http://www.aota.org/en/Education-Careers/Continuing-Education/ForProviders.aspx	Assessment of Capacity for Myoelectric Control
Skills for Life http://www.usispo.org/skills_for_life.asp	RIC Resource Library http://lifecenter.ric.org/ Search Health care topic: medical information & care Subtopic: amputee	Otto Bock http://www.ottobockus.com	US-ISPO www.usispo.org and International ISPO http://www.ispoint.org/ispo-elearning	Box & Blocks
AAOP http://www.oandp.org/	Art Heinze Bilateral Upper Limb Loss www.armamputee.com	RSL/Steeper http://www.rslsteeper.com/	AAOP (interactive webinars, online tutorials, conferences) http://www.oandp.org/olc/	Activities Measure for Upper-Limb Amputees

AOPA www.aopanet.org/	Eric Nelson: Bilateral Upper Limb Loss ednelson@fgn.net	Texas Assistive Devices www.n-abler.org	Clinical Education Concepts (live courses) http://www.cecpo.com/	Jebsen Taylor Test of Hand Function
AOTA www.aota.org	YouTube Search Keywords: Arm amputation Arm prosthesis Upper limb loss rehabilitation Upper limb loss Upper limb prosthesis Upper extremity prosthesis Bilateral upper limb loss rehabilitation Arm prosthetic device	Touch Bionics http://www.touchbionics.com/	Various prosthetic care facilities provide in-services or continuing education events Find local providers at http://www.abcop.org	Official Findings of the State of the Science Conference #9: Upper Limb Prosthetic Outcome Measures (AAOP Publication) http://www.oandp.org/jpo/library/index/2009_04S.asp
Amputee Coalition http://www.amputee-coalition.org	—	TRS, Inc http://www.oandp.com/products/trs/about/	—	Clothespin Relocation Test
—	—	Vincent Prosthetic Systems http://vincentsystems.de/en/	—	—
—	—	*This list does not include prosthetic patient care facilities that may make or be developing their own devices or products that are not commercially available	—	—

Abbreviations: AAOP, American Academy of Orthotists & Prosthetists; AOPA, American Orthotic & Prosthetic Association; AOTA, American Occupational Therapy Association; ISPO, International Society for Prosthetics and Orthotics; RIC, Rehabilitation Institute of Chicago.

the electrodes, computer software, and prosthetic hardware so they can feel what it is like and what is involved when operating a prosthetic device. Some components (hands, wrists, elbows) can be controlled proportionally, which means clients have the ability to vary the speed and/or the pinch force based on how hard or soft they contract their muscle. Other components can only be controlled with one speed despite the strength of their muscle contraction, which is often referred to as *digital control*. Some prosthetic systems have the ability to perform simultaneous control of 2 or more components (for instance, simultaneous movement of the elbow and the terminal device), whereas others require clients to switch between components by eliciting a certain muscle signal or using an external physical switch. There are several options to choose from when switching between components, and they can be tested out virtually or during prosthetic training to see which modes clients perform better with. Two of the most common modes of switching are typically referred to as *quick/slow* and *cocontraction*.

The quick/slow method requires clients to send a quick and strong muscle signal within a certain time frame. For example, if clients reach the preset threshold with their extensor muscle group within the time frame, the wrist may supinate. If their muscle signal does not reach the preset threshold within that time frame, then the terminal device will open. The cocontraction method requires clients to contract both muscles at the same time, also within a certain time frame, to move from component to component (elbow, wrist, or hand). For example, if clients are currently operating their elbow and they want to now operate their hand, they must cocontract both muscles in order to switch to their hand so they may control it. There are other methods to switch between components, although they are less commonly used and typically require more extensive training. Terminal devices, in particular, may have multiple programs to choose from and give clients the option of operating the device the best way that suits their needs (proportional, digital, 1 muscle site or 2 muscle sites). Some terminal devices also have the ability to perform an automatic grasp (detects when a grasped object is slipping and tightens pinch force to avoid losing grasp) or pulsing grasp (unique pulsing for increasing grip force, selectable by digit). It is important for the OT to know how the prosthesis is controlled and which programs are chosen for their clients so they can provide the appropriate training.

Electric terminal devices have been commercially available since the 1960s. The technology remained fairly unchanged until the 1990s with the introduction of proportional control and then in the 2000s with the introduction of multi-articulating hands. Many health care professionals are familiar with the standard tripod grasp that has been used for decades for the grasp and release of objects. Prosthetic manufacturers have recently introduced terminal devices that provide the following:

- Different hand positions
- Flexible wrists
- Digits that move independently
- Individual digits for those missing one or more fingers

With these new components, clients are now able to achieve various hand positions that were not possible before:

- Lateral pinch
- Index point
- Finger abduction/adduction
- Other customizable positions

For those clients who are familiar with the standard tripod pinch technology, these multi-articulating terminal devices may require retraining so they can take full advantage of the new features instead of just relying on the old features. For clients who have not used a previous electric prosthesis, the training may be more straightforward and retraining will not be necessary. Please see the manufacturers' Web sites (see **Table 2**) for more detailed information on specific training tools for their devices. It is also a good idea to get to know your local prosthetic manufacturer representative because they will always have the most up-to-date information.

COMMUNITY REINTEGRATION AND OTHER REFERRALS

In addition to the prosthetic training aspect, the OT should also be involving clients in community-reintegration activities:

- Grocery store runs
- Eating out at a restaurant
- Pumping and paying for gas
- Post office errands
- Working out
- Other daily errands/activities

This practice, with the aid of a skilled health care provider, will provide valuable lessons on how to approach these tasks in a more efficient and body-mechanic-friendly way as well as increase the clients' level of confidence in performing activities on their own. Other areas the OT should address in therapy or by requesting referrals from the physician include, but are not limited to, the following:

- Worksite evaluation
- Home evaluation
- Functional-capacity evaluation or work-hardening evaluation
- Return-to-driving evaluation
- Mental health referrals
- Vocational rehabilitation
- Assistive technology

MULTI-LIMB LOSS

Working with someone who is missing multiple limbs presents unique challenges. There are 6 major points for an OT to keep in mind:

- First, barring any major issues, the longer side is typically treated as the dominant side.
- Second, prosthetic training should begin with the dominant side.
- Third, training should happen one component at a time:
    - Should begin with the terminal device because it is most often used
    - Move on to the next, most-used component (the elbow if transhumeral and the wrist if transradial)
- Fourth, training should progress to the following:
    - Bimanual controls training
    - Repetitive drills
    - Two-handed skill training (see the section on the 3 stages of prosthetic training)
- Fifth, adaptive equipment is a must for someone with bilateral upper limb loss.

Individuals with multi-limb loss should be shown how to accomplish basic self-care tasks with and without their prostheses. In the event they are without their arms for any period of time, it is essential in helping them maintain a certain level of independence. Adaptive equipment will vary from person to person depending on their level of limb loss and desires for independence in the home. For higher levels of limb loss, a helpful chart with adaptive equipment considerations and resources is provided (ie, a dressing tree is extremely beneficial for independence with dressing and don/doff of prostheses or a bidet for independence with toileting) (Table 3).

- Sixth and last, a home evaluation is essential to evaluate what areas of the home may need modification in order for clients to be independent with self-care or home-management tasks. Working with a skilled and knowledgeable Americans with Disabilities Act residential contractor is extremely helpful. See Fig. 5 for a shower set-up example.

Table 3
Bilateral upper limb loss adaptive equipment considerations

ADL Area	Equipment
Toileting	Bidets with wall-mounted control, travel variety, wet wipes, toilet paper on base of toilet
Feeding	U cuffs, swivel utensils, Dycem® (Dycem Ltd, Bristol, UK)
Dressing Prosthetic don/doff	Dressing Tree, industrial strength suction cups with hooks
Grooming, feeding	Goose neck clamps or industrial-strength suction cups with hooks or with adaptive holder on end (hair dryer, razor, deodorant, brushing teeth, suction cup scrub brush, wall-mounted brush)
Communication/computing	Home phone: speakerphone with large buttons Computer, tablet, or smart phone: voice recognition software, mouth stick
Environmental controls	High-technology computerized control: lights, thermostat, kitchen appliances, caregiver call, TV, computer Basic control: mouth stick
Bathing	Wall-mounted sponges, brushes, or soap/shampoo dispenser, handles instead of knobs, pedals for dispensing, prosthetic shower arms, body dryer
Touch screens (phones, tablets, thermostat, kiosks, banking)	Capacitive vs resistive touch screens: specialized gloves (REI/Target), packaged beef jerky, packaged string cheese, specialized stylus, O-Cel-O™ sponge (3M, St. Paul, MN, USA)
Female considerations	Bras, monthly cycles, makeup, hair, shaving legs
Meal preparation	Jar/can opener, paring board, pot/dish holder, dycem, pouring aid, water or beverage dispenser, one-touch storage containers, easy-access condiments
Around the house	Rocker switch lights, handles on doors, swinging doors
Driving	Adapted controls, foot controls or head rest controls for steering, air conditioning/heat, stereo, windows, ignition, horn, wipers, lights, blinkers
Keep in mind preferences, weather/climate	Elastic pants vs jeans, slip-on shoes vs shoelaces, Velcro (Velcro Industries, Manchester, NH) vs buttons or zippers

Fig. 5. Bilateral shower setup. (*Courtesy of* Brooke O'Steen, Fort Wayne, IN.)

For those individuals without one or both legs, the adaptive equipment needs and home modifications will vary and be more involved.

OUTCOME MEASURES FROM THE OT'S PERSPECTIVE

There are various standardized tests that OTs use in a clinical setting to document baseline status and progress to prove therapeutic intervention is medically necessary. However, many of these commonly used tests are not validated for use with someone who has a prosthesis. Some tests are just not feasible secondary to the length of time to complete and the cost of the test or training to administer the test. Some peer-reviewed articles mention slight modifications to certain tests to accommodate the prosthesis, but these results are not validated and conducted on such a small sample size that it is not meaningful. For more detailed information on the state of upper limb prosthetic outcome measures, please see article by Heinemann and colleagues elsewhere in this issue.

There are 4 outcome measures this OT recommends for use in a clinical setting (see Table 2).

1. The Assessment of Capacity for Myoelectric Control
2. Southampton Hand Assessment Procedure
3. Box and Blocks
4. Jebsen Taylor Test of Hand Function

There are many other tests that can also be used depending on what question you are trying to answer and what component of the International Classification of Functioning, Disability, and Health you are addressing (function, activity, participation).

RESOURCES

Typically, an occupational therapy student will spend a few hours in school reading a few pages in a text book about upper limb loss rehabilitation. When practicing OTs do encounter someone missing an arm, they may only see one every few years, which can be attributed to the sheer low numbers of individuals with upper limb loss. This area of rehabilitation for an OT is greatly lacking in continuing education and useful resources. It can become a highly specialized field for an OT who works with these individuals on a regular basis. This OT recommends connecting with prosthetic care facilities experienced in upper limb loss care and prosthetic manufacturers as well as performing thorough Internet searches to learn about conferences, videos, outcome measures, and specific courses related to prosthetic rehabilitation Please see **Table 2** for suggested resources.

TECHNOLOGY: WHAT IS NEW?

Targeted muscle reinnervation (TMR) is a relatively new elective surgery that increases the number of electromyography (EMG) control signals, which in turn has the potential for enhancing prosthetic function. The typical myoelectric prosthesis most OTs are accustomed to have the ability to use up to 2 EMG signals for control. A TMR myoelectric prosthesis has the ability to use up to 6 EMG control signals. The clients with this surgery were seen in research-and-development settings initially for evaluation and testing, but the surgery is now being done worldwide; clients are now being seen in clinical settings with excellent outcomes. An occupational therapy training protocol was developed and is invaluable when working with a TMR client.[7]

Pattern recognition is another recent development in upper limb prosthetics. During the evaluation phase while being hooked up to a group of surface EMG electrodes and computer software, clients will move their phantom limb into a position, for example, hand open, point index finger, elbow flex, wrist supinate, and so forth. The surface electrodes and computer software will pick up a pattern based on what it reads from the group of muscles doing the movements. The various patterns for the different hand and arm movements can be programmed into a client's prosthesis for the prosthetic training phase. The prosthesis may need to be recalibrated every once in awhile, but it is a quick and easy process. The need for 2 or 6 EMG signals is no longer required, which translates to more degrees of freedom and more simultaneous abilities. This innovation is still mostly in the research phase because the commercially available prosthetic hardware is currently not compatible with the pattern-recognition software.

SUMMARY

The prosthetic training process for individuals with upper limb loss can be lengthy; but with a well-designed plan of care, an involved client, early fitting/training, and a collaborative team approach, it can be a very rewarding experience for all parties involved. Rehabilitation will most likely occur in multiple phases over a period of many years because clients will have different ADL needs over time, changes in health can occur, and new componentry is always being developed.

Successful functional outcomes are attainable, and it is important for the OT to manage expectations, seek out available resources, understand the componentry to determine training needs, and continually assess the abilities of clients with upper limb loss to move on to the next stage of skills.

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